

Forensic Sketch Generator: An AI-Powered Approach

Using Natural Language Processing and Conditional GAN



Report

Science and Technology advancement- Assessment Review (STAAR)

Submitted by:

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ABSTRACT

This project presents an efficient and automated approach for generating forensic face sketches from natural language witness descriptions using **Natural Language Processing (NLP)** and **Conditional Generative Adversarial Networks (cGAN)**. Forensic sketch generation is a critical step in criminal investigations, yet it traditionally relies on skilled sketch artists who are scarce, expensive, and introduce subjective biases. The proposed system addresses these challenges by bridging the gap between verbal witness accounts and visual forensic evidence.

The proposed method operates as a multi-stage pipeline. Witness descriptions are first parsed using an NLP module built on **spaCy and HuggingFace Transformers**, extracting thirteen structured facial attribute categories including face shape, eye colour, hair length, and distinctive marks. These attributes are encoded into a 512-dimensional conditioning vector driving a Conditional GAN based on the StyleGAN2 architecture to synthesise a photorealistic face image. A sketch style transfer module then converts the photorealistic output into a forensic pencil-sketch using edge detection and dodge blending techniques.

The work is grounded in the foundational base paper: Forensic Face Photo-Sketch Recognition using a Deep Learning-based Architecture (Galea & Farrugia, IEEE Signal Processing Letters, 2017) [1], which demonstrated that transfer learning on VGG-Face combined with 3D morphable model data augmentation can bridge the modality gap between sketches and photographs, achieving 94.61% Rank-1 accuracy when fused with LGMS. The proposed system extends this vision by generating sketches from text, creating a fully automated forensic identification pipeline. Performance evaluation demonstrates a FID score of approximately 42.3, facial attribute accuracy of 78%, and identity preservation of 71% at Rank-10.